Shashwat Singh

Portfolio ↗

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Education

Dehradun Institute of Technology — Dehradun, India
Bachelor of Technology in Computer Science and Engineering (CGPA: 8.51) — Aug. 2024 – June 2028

Technical Skills

Languages: C/C++, JavaScript (ES6+), Python, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Express.js, Tailwind CSS, LangChain, Streamlit
Backend & DB: Node.js, Express.js, JWT, Better-Auth, MongoDB, PostgreSQL
Tools & Platforms: Git, Vercel, Ollama
Core Strengths: Full-Stack Web Development, AI/RAG Integration

Projects

Simple AI Assistant | Python, Ollama, Qwen3:4b | **GitHub** — Oct. 2026 – Present

- Developed a command-line AI assistant in **Python** connecting to a locally running model via **Ollama's OpenAI-compatible API** (**Qwen3:4b**) for offline inference, with a rolling **messages history** for multi-turn context.
- Added **function/tool calling** — passing tool schemas to the model, parsing JSON arguments, dispatching to Python functions, and returning results as tool messages; designed a modular tool layer (`tools.py`) with a declarative `TOOLS` schema and name-to-function dispatch map.

AI RAG Document Q&A System | LangChain, Gemini, Streamlit | **GitHub** — Oct. 2026 – Present

- Built a Retrieval-Augmented Generation (RAG) document Q&A app in **Python** using **LangChain** and the **Google Gemini API**, letting users upload PDF/TXT/Markdown files via **Streamlit** and ask natural-language questions grounded in retrieved document context.
- Implemented the ingestion and retrieval pipeline: extracted PDF text with **PyPDF**, chunked with **RecursiveCharacterTextSplitter** (1,000 chars, 200 overlap), embedded chunks with **Gemini embeddings**, and indexed them in LangChain's **InMemoryVectorStore** for semantic search.
- Engineered the RAG prompt flow to retrieve the top-four chunks with source metadata and constrain the model to the provided context; added multi-document upload, conversation history, and automatic re-indexing on file changes.

Certifications

EY & Microsoft, Simplilearn — 2026

- **AI Skills Passport** (EY & Microsoft) | **Intersection of AI and Environmental Sustainability** (EY & Microsoft) | **Introduction to Generative AI Studio** (Simplilearn) | **Microsoft Copilot Studio** | **Data Analyst 101** | **SQL Analytics & BI on Databricks**

Achievements & Hackathons

Smart India Hackathon (SIH) | National Participant — 2024 – 2025

- **2025**: Developed "Namaste API" — an ICD-11 integration layer standardizing AYUSH morbidity reporting across district health pipelines interoperating with national health data systems.
- **2024**: Architected a **3D AYUSH Herbal Realm** prototype leveraging web-native 3D rendering.

Data Structures & Algorithms | LeetCode | **Profile** — 2026 – Present

- Solved over **400+ problems**; proficient in Dynamic Programming, Graph Theory (BFS/DFS, Dijkstra), and tree-based structures.

Leadership & Extra-Curriculars

Member, Debating Society: Refined public speaking and reasoning through inter-college debate circuits.
Interests: Philosophical Literature (Existentialism/Ethics), Competitive Badminton.